Investigation of the aerodynamic properties of an asymmetrically flapping 2D elliptical airfoil

Project-I (NA47005) report

by

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Indian Institute of Technology Kharagpur Autumn Semester 2022-2023

November 30, 2022

DECLARATION

I certify that

(a) The work contained in this report has been done by me under the guidance of my supervisor.

(b) The work has not been submitted to any other Institute for any degree or diploma.

(c) I have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.

(d) Whenever I have used materials (data, theoretical analysis, figures, and text) from other sources, I have given due credit to them by citing them in the text of the thesis and giving their details in the references. Further, I have taken permission from the copyright owners of the sources, whenever necessary.

Date: November 30, 2022 (Sohan Roy)

Place: Kharagpur (19NA10030)

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CERTIFICATE

This is to certify that the project report entitled "**Investigation of the aerodynamic properties of an asymmetrically flapping 2D elliptical airfoil**" submitted by **Sohan Roy** (Roll No. 19NA10030) to Indian Institute of Technology Kharagpur towards fulfillment of requirements for the award of degree of Bachelor of Technology in Ocean Engineering and Naval Architecture is a record of bona fide work carried out by him under my supervision and guidance during Autumn Semester 2022-2023.

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Date: November 30, 2022
Place: Kharagpur

Abstract

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Department: **Department of Ocean Engineering and Naval Architecture**

Thesis title: **Investigation of the aerodynamic properties of an asymmetrically flapping 2D elliptical foil**

Thesis supervisor: **Professor Sunil Manohar Dash**

Month and year of thesis submission: **November 30, 2022**

This article addresses the negative time-averaged thrust performance of a two-dimensional elliptic symmetrical flapping airfoil at very high flapping frequencies. An attempt is made to improve the thrust performance by employing two different flapping frequencies during the downstroke and upstroke of a flapping cycle, thereby making the airfoil asymmetric. For our investigation, we take into account three different pivot locations, three effective angle of attack amplitudes, and a particular range of St, while keeping the Reynolds number constant at 5000. In general, the upstroke flapping frequency is kept constant as the critical Strouhal number, but the downstroke flapping frequency is always chosen with a value higher than the critical Strouhal number. It is important to note that the observations are periodic and appropriately switched when the flapping cycle's frequency selection is reversed. At frequency ranges where the thrust is degrading, asymmetric flapping stroke configurations produce more time-averaged thrust and non-zero lift. Interestingly, when a faster downstroke and slower upstroke configuration is used, the leading edge side pivot point generates positive lift. The centre and trailing edge pivot points operate in the opposite manner. We have further analysed the transient thrust and lift force

profiles and their associated flow structures for all the pivot locations in order to understand these improved aerodynamic performances at high flapping frequencies.

Acknowledgements

I would like to express my sincere gratitude towards my mentor Professor Dr. Sunil Manohar Dash at the Bio-Inspired Aero and Hydrodynamics Laboratory (BIAH Lab), Department of Aerospace Engineering, IIT Kharagpur, who has constantly motivated me to do this insightful and elegant project. He has always taken care of my work to ensure it is executed well. This project was an important experience for me to learn about carrying out extended amounts of research work, and it has been a great privilege and honor to work under his guidance. I am also grateful to Research Scholar, Mr. Jit Sinha, at the BIAH Lab, who guided me at each step of the simulation execution process in great detail which provided me with a golden opportunity of accelerated learning in this field. On a concluding note, I must mention the constant support my family and my peers have provided me in both technical and intellectual aspects which motivated me to accomplish this project.

The project team at BIAHRL is also immensely grateful to **IIT Kharagpur** for computational support, **SERB (**Science and Engineering Research Board), and **ARDB** (Aeronautics Research and Development Board) for financial support.

Contents

Declaration		1
Certificate		2
Abstract		3
Acknowledgements		5
Contents		6
Symbols		7
1	Introduction	8
2	Literature review and objectives	8
3	Problem statement	10
4	Numerical methodology	12
5	Results and discussion	16
6	Conclusion	32
7	Future scope	33
8	Publications	34
Bibliography		35

Symbols

A	*Width of the wake*	*[m]*
c	*Chord length*	*[m]*
C_T	*Thrust co-efficient*	--
C_L	*Lift co-efficient*	--
f_X	*Flapping frequency*	*Hz*
h_0	*Heaving amplitude*	*[m]*
$h(t)_X$	*Instantaneous heaving function*	--
L	*Span of airfoil*	*[m]*
Re	*Reynolds number*	--
St	*Strouhal number*	--
t	*Physical dimensional time*	*[sec]*
T	*Thrust force*	*[N]*
T'	*Physical flapping cycle time*	*[sec]*
U_∞	*Free stream velocity*	*[m/s]*
S	*Planform area of the airfoil*	*[m^2]*
ω_X	*Angular flapping frequency*	*[rad/s]*
$\theta(t)$	*Instantaneous pitching function*	--
θ_0	*Pitching amplitude*	*(°)*
ϕ	*Phase angle*	*(°)*
v	*Kinematic viscosity*	*[m2 /sec]*
ρ	*Density*	*[kg/m^3]*
α_0	Maximum effective angle of attack	*(°)*

Keywords: Pivot point locations; Asymmetric flapping airfoil; Non-zero lift, Water tunnel experiments; Ansys Fluent.

INTRODUCTION

In studies of symmetric flapping airfoils, thrust performance decreases significantly at a high flapping frequency (or St) during forward flight. However, a large number of unmanned aerial vehicles operate in this St range. In addition, for symmetric flapping, the time-averaged lift force generated during a cycle is negligible. This issue can be resolved by introducing asymmetry into the pitching motion by utilising two distinct flapping frequencies for the downstroke and the upstroke

LITERATURE REVIEW AND OBJECTIVES

The effect of asymmetric oscillation of a NACA 0012 airfoil was studied computationally at Re = 1.35105 for a range of oscillation amplitudes and lower frequency in [1]. According to their findings, the asymmetric ratio influences the instantaneous force coefficients and associated flow topologies. It was also discovered that a faster downstroke could delay the production of Leading Edge Vortex (LEV) and improve thrust performance. In a study of a similar nature, [2] numerically examined the influence of asymmetric sinusoidal oscillating motion on the propulsion performance of a pitching foil and attempted to determine if the low thrust generated by pure pitching might be enhanced by asymmetric motion. Results indicate that a greater asymmetry ratio can create a stronger reverse Von Karman vortex in the wake, resulting in increased thrust at the expense of propulsion efficiency. The effect of symmetric and asymmetric pitching foils undergoing sinusoidal oscillation was studied statistically by altering the thickness based on the Strouhal number in [3]. They demonstrated that the time-averaged thrust increases as asymmetry increases. A greater asymmetric constant value accelerates the transition from drag to thrust. In addition to all of these numerical studies, [4] conducted a numerical and experimental analysis of the effect of asymmetric flapping of a tandem-foil system at various phase angles and Re =

5000. Maximum thrust is generated via an increased asymmetric ratio and 180-degree phase difference. In addition, the impacts of the pivot locations on the propulsion performances of two-dimensional flapping airfoils have been examined in a number of relevant studies. Following a ramp-hold type pitch and sinusoidal heave motion, [5] and [6] conducted particle image velocimetry (PIV) and dye visualisation studies on an SD7003 flapping airfoil at different pivot points and flapping

frequency. Changes in the pivot location affect the growth of leading-edge vortices (LEVs) and wake structure, according to the researchers. [7] has expanded [5]'s research in order to investigate the impact of pivot placement on the aerodynamic response of the SD7003 airfoil for unequal pitch and plunge frequencies. By altering the pivot point and flapping frequency, [8] and [9] have undertaken experiments on a pitching airfoil from a different perspective. They discovered that shifting the location of the pitching pivot towards the leading or trailing edge of the airfoil adds a heaving motion component to the actual pitching motion profile. [10] numerically investigated the influence of pivot site on a solely pitching NACA 0012 airfoil and deduced the scaling laws of its propulsive performance (thrust, power, and efficiency) as functions of the pitching axis location and reduced frequency. In addition, a recent computer work [11] has correlated the flow mechanism and aerodynamic response of a sinusoidal pitching airfoil for a variety of fixed and moveable pivot sites. They have developed a new concept of the effective angle of attack to elaborate on the observed flow mechanisms at various pivot points. In contrast to studies of forced flapping airfoils, [12] and [13] examined the effect of pivot locations on the wake regime and mode of vibration of a fully passive flapping airfoil.

PROBLEM STATEMENT

The upstroke and downstroke of a symmetrically flapping elliptic airfoil are depicted in Fig. 1. In this case, the airfoil executes a sinusoidal heaving and pitching motion profile during a whole flapping cycle in accordance with Equations (1) and (2). In symmetrical flapping, the downstroke and upstroke frequencies (f_X) are also identical. In our current research, however, they are distinct. The difference in phase angle (φ) between the heaving and pitching profiles is 90°, as this phase angle produces the best propulsive efficiency [14].

$$h(t)_X = h_0 \cos(2\pi f_X t) \tag{1}$$

$$\theta(t)_X = \theta_{0X} \cos(2\pi f_X t - \phi) \tag{2}$$

h_0 = heaving amplitude = 0.75c for both the downstroke and the upstroke has been formulated in the foregoing equations. However, depending on the downstroke and upstroke frequencies, θ_{0X} = pitching amplitude will change. The chord length of the flapping airfoil is c = 0.04m. The incoming freestream flow velocity U_∞ has been set to 0.125 metres per second. f_X and ω_X represent the flapping frequency and angular frequency of flapping, respectively. In order to determine if it is possible to mitigate the thrust reduction at a high flapping frequency, we coupled two flapping frequencies in this study: the thrust reduction frequency and the frequency at which maximum thrust is obtained at a given effective angle of attack amplitude (α_0). The thrust and lift force performance of an asymmetric flapping airfoil is investigated at three pivot point locations, A, B, and C, at distances 0.25c, 0.5c, and 0.75c from the leading edge, respectively. The pivot points are depicted schematically in Figure 1.

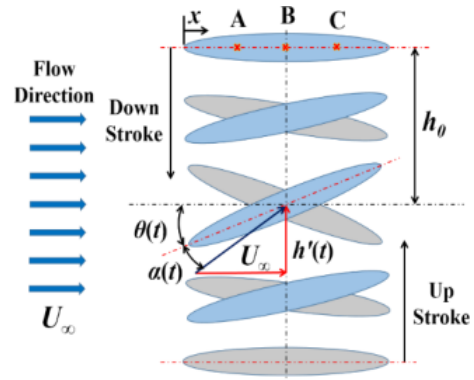

FIGURE 1: SHOWS THE FLAPPING CYCLE OF THE SYMMETRICALLY ELLIPTIC AIRFOIL. THE PIVOT POINTS A, B, AND C ON THE AIRFOIL ARE LOCATED AT 0.25c, 0.5c, AND 0.75c, RESPECTIVELY FROM THE LEADING EDGE.

We also changed the effective angle of attack amplitudes as 10°, 15°, and 20° to evaluate its effects on the propelling performance of the asymmetric flapping airfoil. For our analysis, we define two non-dimensional numbers: the Reynolds number, $Re = U_\infty c/\nu$, and the Strouhal number, $St_X = f_X A/U_\infty$. As f_X differs in downstroke and upstroke, and so does the related St_X. Here, ν is the fluid's kinematic viscosity and A is the wake width (= maximum trailing edge excursion, or $2h_0$). Where T is the thrust force, ρ is the fluid density, and S is the planform area of the airfoil, the transient and time-averaged thrust coefficients followed in the present work are stated in Equations (3) and (4), respectively.

T' represents the flapping cycle time. Similarly, the transient and time-averaged C_L can be defined by substituting the thrust force with the appropriate lift force for the flapping cycle.

$$C_T = \left(\frac{T}{0.5\rho U_\infty^2 S} \right). \quad (3)$$

$$\overline{C_T} = \frac{1}{T'} \int_0^{T'} C_T \, dt. \quad (4)$$

NUMERICAL METHODOLOGY

In the present study, ANSYS FLUENT is utilised to simulate the transient incompressible flow over a flapping elliptic airfoil at Re = 5000 and varying St with a laminar viscous flow model. The efficacy of the ANSYS FLUENT solver in simulating flapping airfoil problems has been fully demonstrated and documented in the literature [16] and [17]. Using the dynamic mesh concept and the arbitrary Lagrangian-Eulerian (ALE) formulation, Eq. (5) is solved in an inertial frame of reference on an arbitrary control volume cv with a moving control surface cs [18].

$$\frac{\partial}{\partial t}\int_{cv}\rho\psi dV+\int_{cs}\rho\mathbf{n}\cdot\left(v-v_g\right)\psi dS-\int_{cs}\rho\Gamma_\psi\mathbf{n}\cdot\nabla\psi dS \quad =\int_{cv}S_\psi\left(\psi\right)dV \qquad \textbf{(5)}$$

where v, v_g, Γ_ψ, S_ψ, and ρ represent the flow velocity, the moving mesh velocity, the diffusion coefficient, and the source term of per unit volume, respectively. n represents the normal unit vector to the control surface. The Pressure-Implicit with Splitting Operators [18] (PISO) algorithm is utilised to couple pressure and velocity. The under-relaxation factors for momentum and pressure are 0.3 and 0.7, respectively. The spatial and temporal discretization of the momentum equation is performed using the second-order upwind scheme and the first-order implicit scheme [18], respectively. To approximate the computer-generated round-off errors, a double-precision solver option is utilised. $O(10^{-5})$ is used as the convergence criterion for both momentum and mass residuals in the iterative technique (10-5).

5.1 Computational domain, mesh and solver validation

In the present simulations, an O-type computational domain is utilised, as depicted in Figure 2(a). The domain is separated into two zones, with the moving inner zone ($\Omega 1$) and the stationary outer zone ($\Omega 2$). There is an interface boundary condition between the two zones. The inner (Γ_1) and outer (Γ_3) boundaries radii are 15c and 30c, respectively, therefore the far-field boundary effects on the airfoil (Γ_0) are insignificant. At the beginning of the flapping stroke, the geometric

centre of an elliptic airfoil is 0.75c from the domain's origin (O). The left half of the outside boundary (2) represents the inlet, while the right half represents the outlet. A uniform freestream velocity (U_∞) in the X direction is allocated to the inlet portion. Conditions of no slip velocity on the airfoil surface and atmospheric pressure on the outlet boundary are respectively implemented. Notably, the two junction locations where the inlet and outflow coincide are positioned a considerable distance from the airfoil surface. Therefore, it has a negligible impact on the near wake flow regime. Utilizing standard initialization, the input result is computed. To achieve the flapping motion profile indicated in Equations (1) and (2), a C-programming language user-defined function (UDF) is designed and compiled in the ANSYS FLUENT solver.

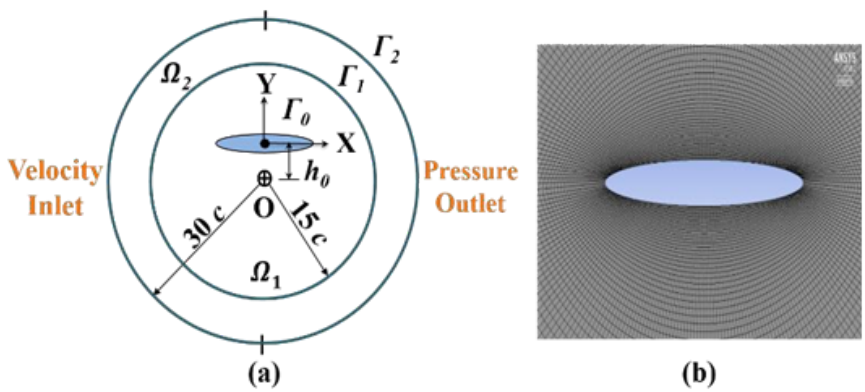

FIGURE 2: (a) REPRESENTS O-TYPE COMPUTATIONAL DOMAIN AND (b) CLOSE-UP VIEW OF THE STRUCTURED MESH NEAR THE ELLIPTIC AIRFOIL.

In this work, a fine mesh with quadrilaterally structured elements is utilised. Figure 2(b) depicts a close-up view of the mesh surrounding the airfoil. Inside the deforming zone, the diffusion smoothing-based dynamic mesh approach was applied. As indicated in Table 1, five distinct mesh sizes were chosen to conduct the mesh independence investigation, with the pivot location kept at the geometric centre of the airfoil (pivot B) and the parameters St = 0.32, Re = 5000, $\varphi = 90°$, $h_0 = 0.75c$, and $\theta_0 = 30°$. The comparison of the transient C_T plots in Figure 3(a) reveals no significant fluctuation with increasing mesh density. In addition, the error rate of time-averaged C_T is less than 0.1 % for mesh size M4. Consequently, this mesh size was used for the remaining simulations in this investigation.

We have analysed a NACA 0012 airfoil undergoing sinusoidal heaving and pitching motion in order to evaluate the present numerical solver as described in Equation (1) and (2). The pivot point is placed at a distance of one-third the chord length (c) from the leading edge. Here, c = 0.01m. As mentioned in section 5.1, the O-type computational domain and the identical set of boundary conditions are utilised. The flapping settings are adapted from [19]; Re = 4000, $\varphi = 90°$, $h_0 = 0.75c$, and St is adjusted between 0.2 and 0.6 with a 0.1-step interval. Here, $\alpha_0 = 20°$, and θ_0 values are calculated according to the processes outlined in [15]. In Figure 3(b), the present solver's transient C_T findings are compared to those of [19]. They are determined to be compatible. This further demonstrates the solver's precision.

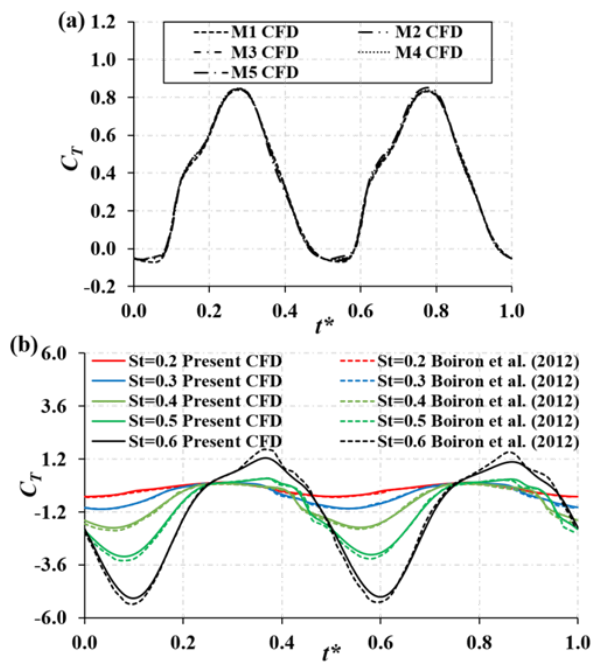

FIGURE 3: (a) SHOWS THE TRANSIENT C_T PROFILE COMPARISON FOR THE MESH INDEPENDENCE STUDY. THE TRANSIENT C_T PROFILE IN THE PRESENT SOLVER IS COMPARED IN (b) WITH THAT REPORTED IN [19].

TABLE 1: MESH INDEPENDENCY TEST

Grid name	***No of nodes***	***C_T value***	***%Error*** $= \left\lvert \frac{CTnew - CTold}{C_T old} \right\rvert \times 100$
M1	29,040	0.3636	-
M2	50,960	0.3659	0.63
M3	**66,300**	**0.3663**	0.11
M4	90,090	0.3666	**0.08**
M5	1,12,230	0.3668	0.05

Results and discussion

We have analysed the thrust and lift performance of the asymmetrically flapping airfoil associated near wake flow characteristics at three distinct pivot locations ('A', 'B', and 'C'; see Fig. 1), three effective angle of attack amplitudes ($\alpha 0$), and a range of non-dimensional flapping frequencies (St = 0.4 – 0.8) in this section.

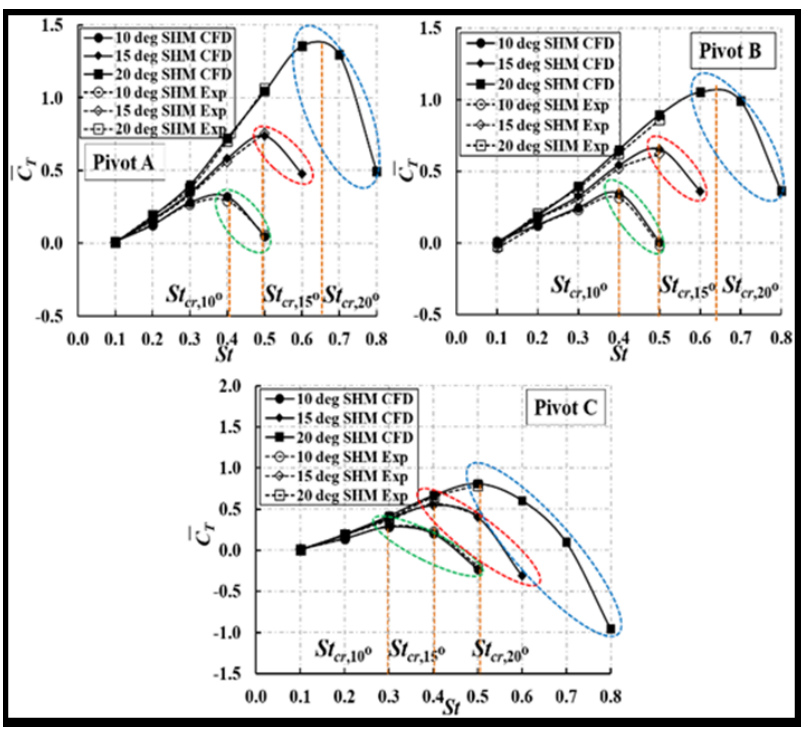

FIGURE 4: NUMERICAL AND EXPERIMENTAL TIME- AVERAGED C_T DISTRIBUTION AS FUNCTIONS OF *St* AND α_0 FOR PIVOT POINT A, B AND C. THRUST REDUCING REGIMES ARE MARKED IN DOTTED CIRCLES. CORRESPONDING *Stcr* ARE ALSO SHOWN BY DOTTED VERTICAL LINES.

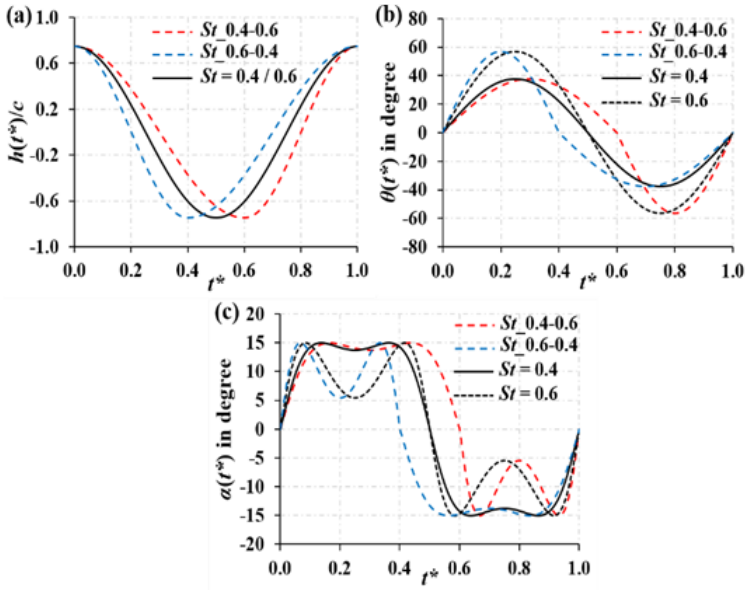

FIGURE 5: INSTANTANEOUS HEAVING, PITCHING AND EFFECTIVE ANGLE OF ATTACK PROFILES FOR THE SYMMETRIC FLAPPING AND COMBINED FREQUENCY CASES ARE SHOWN IN (a), (b) AND (c).

This is the range of St where thrust performance begins to deteriorate at the chosen α_0 values for the symmetric flapping elliptic airfoil [15, 16, 17]. Figure 4 depicts the numerical and experimental time-averaged C_T distribution as a function of St and α_0 for pivot points A, B, and C. The thrust-reduction regimes are denoted with dashed circles, which represent the parametric space of our analysis.

To achieve increased time-averaged thrust at high flapping frequencies ($St > St_{cr}$) for a given α_0, we have implemented two distinct flapping frequencies during the downstroke and upstroke of the flapping cycle. Specifically, the downstroke flapping frequency is always chosen to be greater than the St_{cr}, whereas the upstroke flapping frequency is fixed at the Strouhal number. It is essential to note that the sequence of St is swapped to provide optimal aerodynamic performance. For instance, if the downstroke and upstroke frequencies are 0.4 and 0.6, respectively, it would be represented as St 0.4 - 0.6. If we exchange the frequencies, the result will be St 0.6 – 0.4.

Figure 5 depicts the heaving, pitching, and effective angle of attack profiles of the present asymmetric flapping airfoil for the symmetric flapping frequencies and a combination of two

frequencies. For a representative case, we chose St = 0.4 and 0.6 as symmetric cases and St 0.4 – 0.6 and St 0.6 – 0.4 as combined frequency cases, while holding α_0 constant at 15°. t* is the dimensionless time scale. t* (=t/T') is the ratio between the computational time 't' and the physical flapping cycle time T'. Here, 't' and 'T' are modified so as to achieve a smooth stroke reversal in cases of combined frequency. Note that the St of the flapping airfoil is altered while the maximum magnitude of the effective angle of attack (α_0) remains constant. Particularly, we have considered three α_0, namely 10, 15, and 20 degrees; the corresponding θ_0 is calculated by following the steps outlined in [17] for a constant h_0 (= 0.75c) (refer to Fig. 6(a)). It is worth noticing that, in Fig. 6(c), at high St (for example, St = 0.6), higher-order sub- harmonics appear in the sinusoidal $\alpha(t^*)$ profiles. Additionally, from Fig. 6(b), it can be interpreted that at a high St value, to keep α_0 constant, a higher θ_0 or higher rate of airfoil rotation is needed. Previous studies such as [17] and [15] reported that this phenomenon generates an adverse suction region near the flapping airfoil, which explicitly degrades its overall thrust performance. So, combining a lower flapping frequency with the higher one during a flapping cycle can mitigate the thrust degradation by reducing or eliminating the effects of a high pitching rate or higher order sub-harmonics. Consequently, a significant amount of thrust enhancement can be obtained along with non-zero lift performance.

6.1 Time-averaged thrust and lift performances

The subsequent section examines the aerodynamic thrust and lift force performance of the two-dimensional flapping elliptic airfoil as a function of the pivot locations (A, B, and C), St, and α_0 values. As demonstrated in Figure 4, the critical St and thrust degradation regimes varied based on the effective angle of attack amplitudes for each pivot point. We mixed two frequencies in order to improve thrust performance at such high St. One frequency is chosen from the thrust reduction regime, while the other is maintained at the critical St, where the highest thrust has been documented for symmetric flapping situations. The time-averaged C_T and C_L values of cases with combined frequency are compared to those of cases with symmetric flapping. Notably, the observations are periodic and appropriately exchanged when the flapping stroke frequency selection is reversed during the flapping cycle.

As a result, the time-averaged thrust remains the same for both frequency combinations, however the time-averaged lift becomes negative in some frequency swapped circumstances.

TABLE 2: TIME-AVERAGED THRUST AND LIFT VALUES FOR SYMMETRIC (St = X) AND ASYMMETRIC (St_X-Y) FLAPPING MOTION AT DIFFERENT PIVOT LOCATIONS AND *α0* VALUES. HERE, X AND Y ARE DIFFERENT FREQUENCIES.

	St	C_T	C_L
Pivot A, 10deg	St = 0.5	0.0466	0.0011
	St = 0.4	0.3190	0.0024
	St_0.5 - 0.4	0.2411	0.0913
	St	C_T	C_L
Pivot A, 15deg	St = 0.6	0.4779	0.0005
	St = 0.5	0.7396	0.003
	St_0.6 - 0.5	0.6571	0.2080

Pivot A, 20deg	***St***	C_T	C_L
	St = 0.7	1.2959	0.0024
	St = 0.6	1.3586	0.0067
	St_0.7 - 0.6	1.3162	0.2320
Pivot A, 20deg	***St***	C_T	C_L
	St = 0.8	0.4939	0.0078
	St = 0.6	1.3512	0.0049
	St_0.8 - 0.6	1.2934	0.5730
Pivot B, 10deg	***St***	C_T	C_L
	St = 0.4	0.3455	0.0018
	St = 0.5	0.0066	0.0011
	St_0.4 - 0.5	0.1515	0.0270

Pivot B, 15deg	***St***	C_T	C_L
	St = 0.5	0.6539	0.0023
	St = 0.6	0.3604	0.0029
	St_0.5 - 0.6	0.5023	0.0220
Pivot B, 20deg	***St***	C_T	C_L
	St = 0.6	1.0531	0.0037
	St = 0.7	0.9901	0.0058
	St_0.6 - 0.7	1.0212	0.2570
Pivot B, 20deg	***St***	C_T	C_L
	St = 0.6	1.0531	0.0073
	St = 0.8	0.3633	0.0089
	St_0.6 - 0.8	0.7812	0.2020

Pivot C, 10deg	***St***	C_T	C_L
	St = 0.3	0.2831	0.0089
	St = 0.5	-0.2330	0.0047
	St_0.3 - 0.5	0.1183	0.4462
Pivot C, 15deg	***St***	C_T	C_L
	St = 0.3	0.2833	0.0157
	St = 0.4	0.2021	0.0071
	St_0.3 - 0.4	0.2200	0.1987
Pivot C, 20deg	***St***	C_T	C_L
	St = 0.4	0.5408	0.0155
	St = 0.5	0.4044	0.0023
	St_0.4 - 0.5	0.4122	0.2066

Pivot C, 20deg	***St***	C_T	C_L
	St = 0.4	0.5408	0.0066
	St = 0.6	-0.3100	0.0037
	St_0.4 - 0.6	0.0789	0.2672
Pivot C, 10deg	***St***	C_T	C_L
	St = 0.5	0.8061	0.0079
	St = 0.6	0.5257	0.0014
	St_0.5 - 0.6	0.5989	0.3035
Pivot C, 15deg	***St***	C_T	C_L
	St = 0.5	0.8061	0.0073
	St = 0.7	0.0929	0.0098
	St_0.5 - 0.7	0.4405	0.0103

	St	C_T	C_L
Pivot C, 20deg	St = 0.5	0.8062	0.0187
	St = 0.8	-0.9600	0.0122
	St_0.5 - 0.8	0.6251	0.0984

In Table 2, only those sets of values for which a positive time-averaged lift is seen are displayed.

According to [17], the time-averaged C_T value for pivot A is always greater than pivot B and C for every given symmetrical flapping frequency and α_0 value. The time-averaged C_L value is determined to be insignificant in all cases of symmetric flapping. The critical St for pivots A and B is comparable; thus, their parametric investigation space is identical. However, pivot C differs due to the early loss of thrust. Comparing the values obtained at St > St_{cr} with the time-averaged C_T acquired by merging two frequencies in a flapping cycle is demonstrated in Table 2 for the whole parametric space explored here. Nonetheless, the time-averaged C_T values are marginally lower than those of St_{cr}. Even if the loss of thrust cannot be delayed at higher St, it is possible to generate greater thrust magnitudes at high flapping frequencies. In addition, this asymmetric flapping motion produces a non-zero time-averaged lift. For pivot A, it has been demonstrated that a combination of a faster downstroke and a slower upstroke improves aerodynamic performance. For pivots B and C, however, a slower downstroke and a faster upstroke are preferable. Perhaps this is why insects, birds, and unmanned

The hinge point of wing-flapping vehicles is located on the front edge. Because more thrust and non-zero lift are created during the faster downstroke and slower upstroke at high flapping frequencies. Variations in the time-averaged and transient aerodynamic force coefficients are

presented for a periodic and convergent flapping cycle. There is no combined St that all three pivot points share. In order to comprehend the aforementioned time-averaged force variations, we have chosen a representative case of St 0.4 – 0.6, $\alpha_0 = 15^0$, and pivot B.

6.2 Time-averaged and transient aerodynamic forces for center pivot location (pivot B)

TABLE 3: TIME-AVERAGED THRUST AND LIFT VALUES FOR SYMMETRIC (St = X) AND ASYMMETRIC (St_X–Y) FLAPPING MOTION AT PIVOT B AND α_0 =15^0. HERE, X AND Y ARE DIFFERENT FREQUENCIES.

	St	C_T	C_L
Pivot B, 15deg	St = 0.4	0.5427	0.0027
	St = 0.6	0.3604	0.0005
	St _0.4 - 0.6	0.4659	0.2812
	St	C_T	C_L
Pivot B, 15deg	St = 0.6	0.3604	0.0127
	St = 0.4	0.5427	0.0011
	St_0.6 - 0.4	0.4622	-0.2735

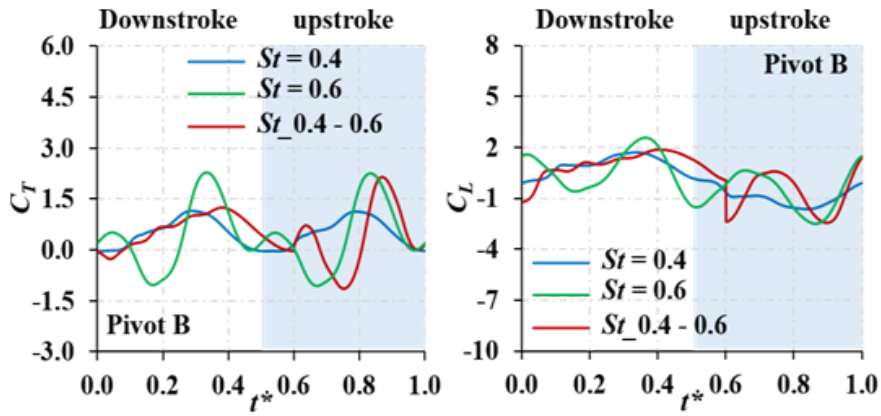

FIGURE 6: SHOWS THE COMPARISON OF TRANSIENT C_T AND C_L PROFILES BETWEEN *St* = 0.4, 0.6 AND *St*_0.4 – 0.6 CASE FOR PIVOT B.

To comprehend the distribution of time-averaged aerodynamic forces, we plotted transient C_T and C_L profiles for the above-mentioned representative case of St 0.4 – 0.6 at α_0 = 15^0 and compared them with the St = 0.4 and St = 0.6 cases in Fig. 7. During the downstroke of St 0.4 – 0.6, the thrust-reducing valley region of St = 0.6 disappeared; however, it persisted during the upstroke, resulting in a thrust value that is greater than St = 0.6 but less than St = 0.4. At St = 0.4 or 0.6, the time-averaged lift provided by symmetric flapping motion is nearly nil. However, for St 0.4 to St 0.6, the transient lift profile generates a non-zero lift, as seen in Table 3. Importantly, when the frequencies in the downstroke and upstroke were reversed, i.e., St 0.6 – 0.4, the transient thrust profile was identical to the St 0.4–0.6 case, the time-averaged thrust
remained unchanged. However, the modified transient profile at St 0.6 – 0.4 results in the same magnitude of negative time-averaged lift (refer to Fig. 8) as the St 0.4 – 0.6 case. Therefore, for the pivot B, a slower downstroke and a quicker upstroke are advantageous for improved aerodynamic performance.

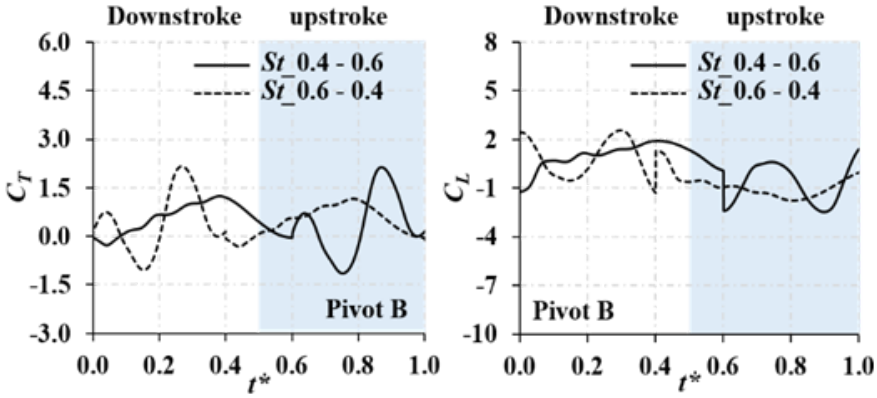

FIGURE 7: SHOWS THE COMPARISON OF TRANSIENT C_T AND C_L PROFILES FOR *St*_0.4 - 0.6 AND *St*_0.6 – 0.4 CASES OF PIVOT B.

6.3 Time-averaged and transient aerodynamic forces for off-center pivot locations (pivots A and C)

To comprehend the effect of pivot point locations on the time-averaged and transient aerodynamic performances, we analysed pivots A and C under the St 0.6 – 0.4 case.

Similar to pivot B, in the case of pivot C also, we can observe a sharp valley region in the transient C_T and C_L profiles near the onset of downstroke (see Fig. 9), which is based on St = 0.6. Incorporating St = 0.4, however, eliminates the valley region on the upstroke. Thus, the time-averaged thrust becomes more than that of St = 0.6, but less than that of St = 0.4. Here also, slower downstroke and faster upstroke are favorable for better aerodynamic performance because, in the case of St_0.6 – 0.4, the deep valley in the transient C_L profile causes negative lift (refer to Table 4). On the contrary, for pivot A, faster downstroke and slower upstroke becomes useful because of upside peak formation in the transient C_L profile (refer Fig. 9 and Table 4)

In terms of thrust and lift performance, pivot A is superior to pivots B and C at any given combined frequency. To comprehend the reason for the superior performance of pivot A, we analysed the transient profiles through pressure and vorticity contours at a specific time t* = 0.25. At t* = 0.25, an extended suction zone encompassed the entire top surface for pivot A, which is

not principally associated with any Leading Edge Vortices (LEVs). Rather, it is the result of Asymmetric Velocity Distribution (AVD) and Previous stroke AVD (PAVD) (refer to [17]). This low-pressure zone created by ADV+PAVD creates significant thrust. Moreover, a high-pressure region on the bottom surface results in superior lift performance. Consequently, significant peaks are created during the downstroke in the transient profiles (observe Fig. 9). At St 0.6 – St 0.4, the smaller LEV on the upper surface and the smaller high-pressure zone on the lower surface of the airfoil create less thrust and negative lift for pivot B. For pivot C, the shedding of LEV from the upper surface and the development of LEV on the lower surface, as depicted in Figure 9, generates relatively little thrust and negative lift forces.

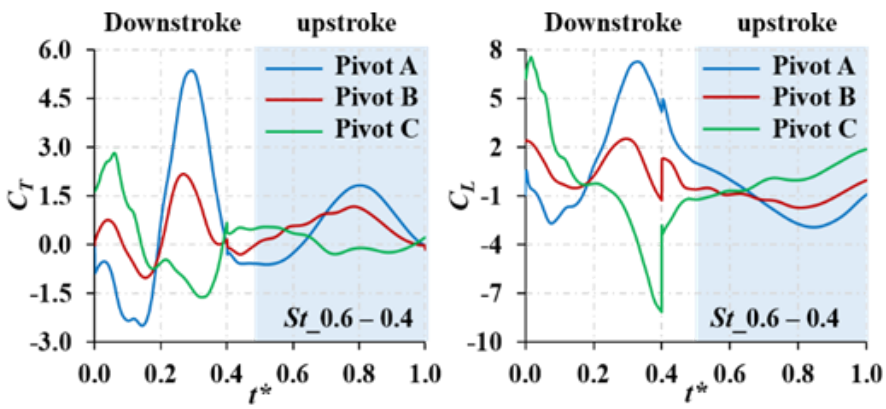

FIGURE 8: SHOWS THE TRANSIENT C_T AND C_L PROFILE COMPARISON FOR *St*_0.6 – 0.4 CASE AT PIVOT A, B, AND C.

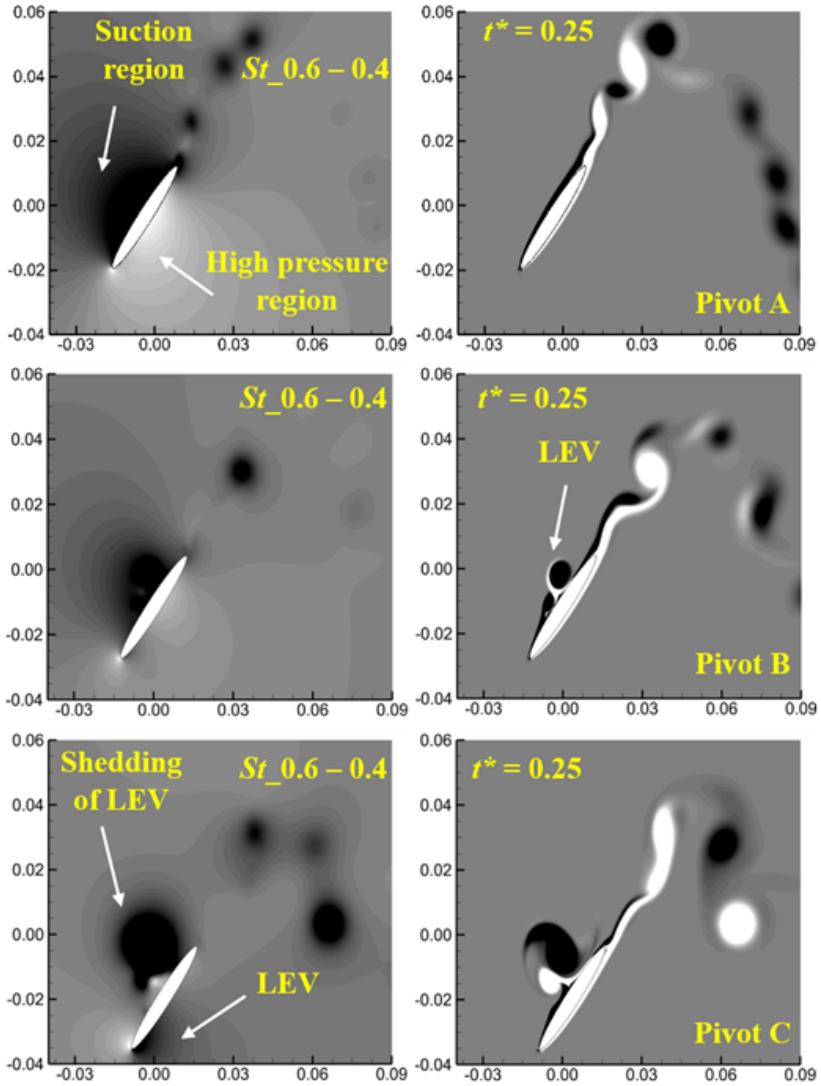

FIGURE 9: SHOWS THE PRESSURE AND VORTICITY CONTOURS FOR *St*_0.6 – 0.4 CASE AT $t^* = 0.25$ FOR THREE PIVOT LOCATIONS (PIVOT A, B, AND C).

TABLE 4: TIME-AVERAGED THRUST AND LIFT VALUES FOR SYMMETRIC (St = X) AND ASYMMETRIC (St_X-Y) FLAPPING MOTION AT PIVOT A AND C FOR α_0 =15^0. HERE, X AND Y ARE DIFFERENT FREQUENCIES.

Pivot A, 15deg	***St***	C_T	C_L
	St = 0.4	0.5827	0.0017
	St = 0.6	0.4779	0.0021
	St _0.4 - 0.6	0.6287	-0.2720
Pivot A, 15deg	***St***	C_T	C_L
	St = 0.6	0.4779	0.0037
	St = 0.4	0.5827	0.0098
	St _0.6 - 0.4	0.6247	0.2833
Pivot C, 15deg	***St***	C_T	C_L
	St = 0.4	0.5408	0.0074

	St = 0.6	-0.3111	0.0099
	St _0.4 - 0.6	0.0795	0.2663
Pivot C, 15deg	***St***	C_T	C_L
	St = 0.6	-0.3100	0.0045
	St = 0.4	0.5408	0.0142
	St _0.6 - 0.4	0.0938	-0.2528

Conclusion

In this numerical study, asymmetric flapping strokes are adopted to mitigate the thrust degradation at high flapping frequency (St) of a symmetric flapping elliptical airfoil in forward flight by implementing two separate St during the downstroke and upstroke of a flapping cycle. Our analysis considers three distinct pivot points at distances of 0.25c, 0.50c, and 0.75c from the leading edge of the airfoil, as well as three α_0 values of 10^0, 15^0, and 20^0. The St ranges between 0.4 to 0.8. In general, the downstroke flapping frequency is greater than the critical St, whereas the upstroke flapping frequency remains constant at the crucial St. The transient and time-averaged data are periodic and appropriately exchanged when the flapping cycle stroke frequency selection is reversed. Asymmetric flapping stroke designs produce greater time-averaged thrust and lift than symmetric flapping stroke patterns. Interestingly, pivot A generates lift with a faster downstroke and a slower upstroke. The opposite is true for pivots B and C. To comprehend these improved aerodynamic performances at high St, we have evaluated the transient thrust and lift force profiles as well as the accompanying flow structures at each pivot site by means of pressure and vorticity contours. Pivot A is superior in all areas for both symmetric and asymmetric flapping instances.

FUTURE SCOPE

1. Experimental verification of the findings through water-tunnel experiments.
2. Analysing the wake flow regimes.
3. Extension of the study to other types of airfoils and flapping kinematics.

PUBLICATIONS

1. Mitigation of Thrust Deterioration at High Flapping Frequencies of a Two-dimensional Elliptic Flapping Airfoil Using Asymmetric Flapping Strokes in the Forward Flight by Sinha J., Roy S. and Dash S. M., 75th Annual Meeting of the APS Division of Fluid Dynamics, USA - (2022)

2. Mitigation of Thrust Deterioration at a High Flapping Frequency of a 2d Airfoil in Forwarding Flight Condition Using Asymmetric Flapping Strokes by Sinha J., Roy S. and Dash S. M., International Mechanical Engineering Congress and Exposition (IMECE2022), USA - (2022)

Bibliography

[1] K. Lu, Y.H. Xie, D. Zhang, Numerical study of large amplitude, non-sinusoidal motion and camber effects on pitching airfoil propulsion, Journal of Fluids and Structures 36 (2013) 184–194.

[2] Q. Xiao, W. Liao, Numerical study of asymmetric effect on a pitching foil, International Journal of Modern Physics Vol. 20 10 (2009) 1663-1680.

[3] L.M. Chao, Md. M. Alam, C. Ji, Drag–thrust transition and wake structures of a pitching foil undergoing asymmetric oscillation, Journal of Fluids and Structures 103 (2021), 103289.

[4] G. Wang, B. F. Ng, Z. W. Teo, K. B. Lua, Y. Bao, Performance augmentation of tandem flapping wings in asymmetric flapping strokes, Aerospace Science and Technology volume 117 (2021) 106939.

[5] M.V. Ol, Vortical structures in high frequency pitch and plunge at low reynolds number, 37th AIAA Fluid Dynamics Conference 2 (2007) 1696–1715.

[6] M.V. Ol, J.D. Eldredge, C. Wang, High-amplitude pitch of a flat plate: An abstraction of perching and flapping, International Journal of Micro Air Vehicle 1 (2009) 203–216.

[7] C. Webb, H. Dong, and M.V. Ol, Effects of unequal pitch and plunge airfoil motion frequency on aerodynamic response, 46th AIAA Aerospace Science Meeting and Exhibit 1–20 (2008).

[8] H.T. Yu, L.P. Bernal, Effects of pivot location and reduced pitch rate on pitching rectangular flat plates, AIAA Journal 55 (2017) 702–718.

[9] W. Tian, A. Bodling, H. Liu, J.C. Wu, G. He, and H. Hu, An experimental study of the effects of pitch-pivot-point location on the propulsion performance of a pitching airfoil, Journal of Fluids

and Structures 60 (2016) 130–142.

[10] R.F. Feria, E.S. Rojas, Effect of the pivot point location on the propulsive performance of a pitching foil, Journal of Fluids and Structures 97 (2020) 103089.

[11] X. Li, L.H. Feng, Z.Y. Li, Flow mechanism for the effect of pivot point on the aerodynamic characteristics of a pitching airfoil and its manipulation, Physics of Fluids 31 (2019).

[12] Z. Wang, L. Du, J. Zhao, X. Sun, Structural response and energy extraction of a fully passive flapping foil, Journal of Fluids and Structures 72 (2017) 96–113.

[13] Z. Wang, L. Du, J. Zhao, M.C. Thompson, X. Sun, Flow-induced vibrations of a pitching and plunging airfoil, Journal of Fluid Mechanics 885 (2020).

[14] J.M. Anderson, K. Streitlien, D.S. Barrett, M.S. Triantafyllou, Oscillating foils of high propulsive efficiency, Journal of Fluid Mechanics 360 (1998) 41–72.

[15] K.B. Lua, S.M. Dash, T.T. Lim, K.S. Yeo, On the thrust performance of a flapping two-dimensional elliptic airfoil in a forward flight, Journal of Fluids and Structures 66 (2016) 91–109.

[16] S.M. Dash, K.B. Lua, T.T. Lim, K.S. Yeo, Enhanced thrust performance of a two dimensional elliptic airfoil at high flapping frequency in a forward flight, Journal of Fluids and Structures 76 (2018) 37–59.

[17] J. Sinha, S.M. Dash, K.B. Lua, Influence of the pivot location on the thrust and propulsive efficiency performance of a two-dimensional flapping elliptic airfoil in a forward flight, Physics of Fluids 33 (2021) 081912.

[18] J.H. Ferziger, M. Peric, Computational Methods for Fluid Dynamics, Springer Berlin, 2002.

[19] O. Boiron, C. Guivier-Curien, E. Bertrand, Study of the hydrodynamic of a flapping foil at moderate angle of attack, Computers and Fluids 59 (2012) 117–124.